

4(a)	product of density and speed	M1
	speed of sound in medium	A1
4(b)	Any two from: - if $Z_A \gg Z_B$ then ratio is (nearly) zero or if $Z_B \gg Z_A$ then ratio is (nearly) zero or if Z_B and Z_A are very different then ratio is (nearly) zero or the greater the difference the lower the ratio - if $Z_A \approx Z_B$ then ratio is (nearly) 1 or if $Z_A = Z_B$ then ratio is 1 or the smaller the difference the closer the ratio to 1 (not 'large') $I_T / I_0 = 1 - [(Z_A - Z_B)^2 / (Z_A + Z_B)^2]$	B2
4(c)	$I = I_0 e^{-\mu x}$	C1
	$0.34 = \exp(-23 \times x)$	C1
	$x = 0.047 \text{ m}$	A1